

This document details the architectural core, system data mechanics, intelligence parameters, and comprehensive interface maps for **TenaxAI**. This framework eliminates information redundancy using active state context propagation, standardizes layout spacing to resolve component congestion, and positions deep high-accuracy frameworks tailored for regional market data parameters.

PART 1: COMPREHENSIVE PRODUCT REQUIREMENT DOCUMENT (PRD)

1. System Architecture & Global Context Engine

1.1 Central Data Schema

The `Project` entity is the definitive platform data layer. Workspace applications, analysis tools, and async processes ingest state data from this synchronized object map. Updates write directly back into the record structure, rendering secondary or redundant manual forms entirely deprecated.

Field Key	Data Type	Functional Context & Injection Scope
product_name	string	Hydrates header components, tool execution labels, and baseline workspace contexts.
description	string	Injected systematically into agent processing structures to establish operational limits.
problem	string	Drives intelligence mapping inside market validation models and alignment panels.
category	string	Acts as core routing mechanism for contextual tool configurations and categorization.
target_markets	string	Generates global structural variables across market, distribution, and design tabs.
target_geos	array[string]	Target country array matching dedicated geographic analysis prioritization logic.
personas_input	string	Hydrates baseline fields for automated persona matrix processing modules.
pricing_info	string	Defines internal data inputs for commercial matrices and modeling loops.
price + currency	num + string	Powers downstream numerical systems, trajectory tools, and predictive computations.
competitors_input	string	Establishes market intelligence and contrast lines for analysis generation.
launch_date	date	Sets chronological alignment intervals and risk timeline tracking benchmarks.
target_budget	number	Sets boundary condition benchmarks for automated simulation processing arrays.
primary_goal	enum	System configuration switch setting operational rules: [Awareness, Leads, Sales, Trial].
document_content	string	Captures detailed asset documents (Truncated at 2,000 characters for background calls).

1.2 Automation & Hydration Rules

- **Rule 1 — Context Extraction:** Selecting a record from the top-level selection panel fires the `getProjectContext(project)` method. Input modules on the screen capture this context and fill data structures instantly to bypass multi-point configuration forms.
- **Rule 2 — Structural State Save:** Manual modifications to fields in specialized workspaces save to the primary schema upon an explicit "Save" action, triggering synchronization across separate execution panels.
- **Rule 3 — Caching Architecture:** Sub-modules cache operational steps in safe browser namespaces using a structured `tenax_{toolKey}_{projectId}` key layout to guarantee state survival across structural transitions.

- **Rule 4 — Injection Context Assembly:** Background calls generate a dense unified context string matching:
Product: {name} | Category: {cat} | Problem: {prob} | Target: {geos} | Price: {pr} {curr} | Budget: {budg}

2. Market Research Core Logic — African Market Intelligence Framework

2.1 Multi-Layered Intelligence Pipeline

When target geographic structures locate execution models in African environments, the platform boots into **African Market Mode**. It processes regional infrastructure matrices across specific operational points:

Operational Filter	Required High-Accuracy Output Dimensions
Mobile & Connectivity	Processes regional connection distributions (2G/3G/4G/5G splits), data overhead realities, mobile operational software footprints, and functional device capabilities.
Payment Infrastructure	Monitors alternative payment platforms (e.g., M-Pesa ecosystem), local fintech layers (Paystack, Flutterwave), regional mobile money usage, banking indexes, and USSD reliance maps.
Platform Intelligence	Evaluates localized professional network activity, tracing dominant usage trends across conversational hubs (WhatsApp Business) and high-velocity channels.
Economic Metrics	Traces relative inflation impacts, employment statistics among target categories, and purchasing power parity data.
Regulatory Context	Assesses compliance requirements including local privacy architectures (such as Nigeria's NDPR or South Africa's POPIA) alongside business setup constraints.
Distribution Frameworks	Details last-mile infrastructure status, distribution hub layouts, and trade ecosystem patterns.

The system applies an intentional calculation logic to output a composite opportunity score (0–100):

Score = (Size * 0.20) + (Payments * 0.20) + (Digital * 0.15) + (Competition * 0.15) + (Regulation * 0.15) + (LocalFit * 0.15)

3. Scenario Simulation Engine Rules

3.1 Simulation Execution Strategy

The simulation engine processes input parameters synchronously via a strict analytical framework designed to deliver concrete predictive projections:

- **Sizing Metrics:** Models standard addressable limits (TAM, SAM, SOM) based on localized market filters and pricing inputs.
- **Adoption Curves:** Employs structured mathematical models to map multi-variant performance tracks over a 12-month timeline, modifying outcomes based on category-specific user stickiness parameters.

- **Financial Metrics:** Generates accurate data models documenting cash parameters, customer access costs, life cycle values, payback durations, and net resource returns.
- **Weighted Profiles:** Distributes outcomes across explicit probability lanes: Conservative (35%), Baseline (40%), High-Growth (15%), and Stress Scenarios (10%).

PART 2: DE-CONGESTION DESIGN SYSTEM & RE-LAYOUT RULES

1. Visual Hierarchy & Space Balancing Rules

To fix workspace layout crowding, the interface applies standard design guidelines to create separation, clarity, and whitespace throughout the screen layout:

- **Layout Architecture:** Standard 12-column layout grid. Maximum layout span restricted to 1280px with centered margin alignment. Outer screen framing sets clear spacing limits (`px-6 py-8` desktop / `px-4 py-6` mobile devices).
- **Spacing Limits:** Main interface blocks maintain strict separation gaps (`gap-6`). Component cards split layout actions evenly (`gap-4`). Individual container interiors apply explicit inner space padding (`p-6` standard).
- **Interactive Boundaries:** Interactive modules maintain minimum safety gaps of 12px. Narrative or text outputs require a minimum 20px clearance boundary away from container framing lines.
- **Container Architectures:** Workspace blocks map to standardized structural specs (`rounded-2xl border border-border/50 bg-white/80 backdrop-blur-sm`). Nested elements are capped at a single layer to avoid complex UI nesting. Divider treatments apply clean line designs (`border-b border-border/40`) solely between block titles and primary interface spaces.

2. Workspace Sidebar Action Rules

- **Expanded View Configuration ("Out"):** Sets standard dimension parameters (`w-64`). Renders brand imagery, title labeling ("TenaxAI"), layout navigation targets, secondary markers, and a complete user profile group. Transitions utilize smooth scaling animations (`transition-all duration-300 ease-in-out`).
- **Collapsed View Configuration ("In"):** Adjusts active dimensions (`w-16`). Hides secondary labels, displaying only primary application icons. Target titles move to independent hover panels.
- **Interface State Trigger:** Interaction toggles map to a prominent navigation icon element. View variables save instantly to local storage via a `tenax_sidebar_collapsed` state property to ensure consistency across active system sessions.

PART 3: SCREEN-BY-SCREEN UI/UX SPECIFICATION

PAGE 1: Project Onboarding & Context Dashboard

System Route: /ProductBuilder | **Role:** Central setup hub for project data orchestration.

TenaxAI Logo

New Launch Workspace

<Badge: Context Hub>

[Save Project]

(1) Basics [Active]

==>

(2) Market [Pending]

==>

(3) Audience

==>

(4) Financials

==>

(5) Docs

SECTION A: CORE BASICS

* Project Name: [PayFlow Central API]

Category: [Fintech v]

System Nature: (X) Digital Architecture () Material Asset

Core Product Description:

[Integrated transaction management engine optimized for processing multi-tier workflows...]

SECTION B: GEOGRAPHIC MAPPING

Target Territories: [Nigeria, Kenya, Ghana, South Africa]

Active Profiles: [[Nigeria x]] [[Kenya x]] [[Ghana x]] [+ Inject Country Profile]

Structural Layout Components:

- 1. **Global Control Banner:** Flexible layout component setting alignment space (flex justify-between items-center mb-8). Houses title typography, workspace tag markers, and the primary Save Project execution trigger.
- 2. **Setup Steps Indicator:** Configures horizontal layout trackers (flex gap-2 mb-8) visualizing progress states across five development checkpoints.
- 3. **Block A — Core Product Basics:** Structured 2-column input grid detailing project name inputs, platform category selectors, deployment rules, and auto-expanding system narrative areas.
- 4. **Block B — Territory Targets:** Mounts explicit country array selectors using interactive badge components to trigger region-specific framework layers.
- 5. **Block C — Commercial & Budget Configurations:** Captures numerical value vectors including transactional values, operating currencies, and launch budgets alongside campaign goals.
- 6. **Block D — Extended Document Context:** Implements a tabbed container enclosing plain-text entry areas and drag-and-drop document parsers to process external files.

PAGE 2: Market & Product Fit Workspace

System Route: /MarketProductFit | **Role:** Validates product positioning, price metrics, and target persona alignment.

[Icon] Market & Product Fit <Pill: PayFlow Central API>										[Project Context v]																			
[Stress Testing]					[Pricing Models]					[Value Parameters]					[Persona Builder (Active)]														
Target User Reference Data: [Core enterprise engineering teams managing ledger connections...]																													
Target Region Mapping: [West Africa Hub - Nigeria v]										[Extract Data] [Export]																			
+-- DETAILED TARGET AUDIENCE GENERATION MATRIX -----+																													
+-----+										+-----+										+-----+									
Persona: Tunde (Product Engineer)										Persona: Amina (Operations Director)										...									
Location: Lagos Tier 1 Device										Location: Nairobi Cloud Infrastructure																			
Pain: API Latency & Settlement Drops										Pain: Interbank Reconciliation Costs																			
Target Focus: WhatsApp / Developer										Target Focus: Direct Field Network																			
+-----+										+-----+										+-----+									
+-----+																													

Structural Layout Components:

1. **Navigation Header:** Displays tracking modules displaying the selected active project profile alongside workspace filtering tools.
2. **Core View Selection Bar:** Implements a clean horizontal toggle line that switches the interface layout across four functional workspaces:
 - **Stress Testing Control Tab:** Renders structural configuration sidebars alongside opportunity matrix dashboards and localized survey question outputs.
 - **Pricing Models Tab:** Hosts transaction tier maps, user sensitivity charts, and competitive layout matrix views.
 - **Value Parameters Tab:** Maps system capabilities directly to user benefit modules, tracking value perception scores on horizontal line scales.
 - **Persona Builder Workspace:** Processes target audience profiles inside a multi-column card layout. Each block isolates customer descriptors, technology requirements, localized workflow problems, and estimated system suitability metrics.

PAGE 3: Creative & Messaging Workspace

System Route: /CreativeMessaging | **Role:** Manages campaign copywriting, localization passes, and channel priority optimization.

Structural Layout Components:

- 1. **Workspace Tab Strip:** Coordinates marketing workflows across four specialized interface views:
 - **Ad Creative Panel:** Generates cross-platform variation grids displaying structured copywriting samples, targeted platform tags, and estimated performance outputs.
 - **Email Architecture Module:** Vertically aligned view space presenting structural communication flows rendered in clean Markdown formatting blocks.
 - **Channel Recommender Dashboard:** Displays primary channel efficiency matrix tables matching investment requirements to expected returns. Includes custom priority visualization blocks, distribution graphs, and infrastructure advice hints tailored for African connectivity channels.
 - **Localization Engine:** Formats parallel interface layouts displaying country conversion tables, target linguistic adaptations, and cultural implementation alerts.

PAGE 4: Scenario Simulation Dashboard

System Route: /RiskSimulation | **Role:** Runs asynchronous modeling threads to simulate financial and adoption trajectories.

```
+-----+
| [Icon] Scenario Simulation Engine (Sub: Context Projection Engine) [Project Context] |
+-----+
|
| +-- SIMULATION MODE SELECTION -----+
| | ( ) Baseline Growth (X) Stress Conditions ( ) Accelerated Case ( ) Disruption Footprint|
| | Timeline Span: [ 12 Months v ] Analysis Horizon: [ Comprehensive Regional View v ] |
| |
| | [===== RUN SIMULATION THREAD (ASYNC PROJECTION) =====]
| |
+-----+
|
| +-- SIMULATION CALCULATION PIPELINE PROGRESS (Visible during background process execution) -----+
| | [===== 78% =====]
| | Current Module Phase: Computing Financial Matrix Profiles... Time: ~6s
| |
+-----+
```

Structural Layout Components:

- 1. **Pre-Flight Validation Alert:** High-visibility information block that displays active system warnings if necessary pricing data fields or target region variables are left incomplete.
- 2. **Simulation Control Interface:** Implements large, selectable radio buttons to configure project simulation types paired with timeline horizon parameters and manual assumption inputs.

3. **Execution Progress Tracking Component:** Active visualization panel that reflects background data calculations. Displays real-time process labels: Loading Context → Sizing Market → Modeling Scenarios → Calculating Financials → Validating Results → Complete .
4. **Analytics Performance View:** Hidden during calculation passes; fades smoothly into view upon compilation. Features key metric summary strips, multi-scenario data matrix sheets, cross-scenario trajectory tracking charts, risk alerts, and export action options.

PAGE 5: Active Resources Directory

System Route: /Resources | **Role:** Consolidated database tracking integration services, compliance architectures, and specialized regional tooling ecosystems.

Structural Layout Components:

1. **Search & Filter Management Strip:** Combines real-time text filter fields with target service categories and location dropdown tools. Includes an explicit action trigger to launch context-aware AI searches.
2. **Vetted Infrastructure Grid:** Multi-column container displaying active tool reference cards. Each block tracks specific taxonomy tags, service documentation clips, direct documentation URLs, and a pulsing live status confirmation indicator to verify external endpoint connectivity.

GLOBAL COMPONENT: Persistent Chatbot Layer

Component Blueprint Target: components/common/ChatWidget.jsx | **Layout Anchor:** Fixed screen positioning anchored to the bottom-right viewport corner.

- **Floating Action Interface:** Standard action node (w-14 h-14) featuring a soft pulsing visual indicator whenever fresh context variables update the underlying system data framework.
- **Interactive Chat Window:** Configures explicit sizing specs (w-[380px] h-[520px]) wrapped in soft container borders and background blurs.
- **Header Controls:** Positions system avatar imagery alongside operational metrics. Integrates a dedicated **Close / Minimize Window Icon Button**.
CRITICAL INTERACTION RULE: Engaging the close action toggles screen visibility tags to a hidden state (isOpen = false) but preserves the message state array. It prevents conversation data loss, maintaining running data sessions across page transitions.
- **Message Streaming Panel:** Scrollable interface containing conversational message balloons: user interactions line up along the right boundary using distinct background colors; AI responses align along the left side displaying markdown-formatted text structures.